KK-L-258a
December 8, 1965
SUPERSEDING
Fed. Spec. KK-L-258
July 27, 1953

FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, SKIRTING, VEGETABLE-TANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cattlehide skirting leather in the form of sides or backs (see 6.1).

1.2 Classification.

1.2.1 *Types and classes.* The skirting leather shall be of the following types and classes, as specified (see 6.2):

Type I—Sides.
Type II—Backs.
Class 1—Mildew resistant treated.
Class 2—Untreated.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

KK-L-311—Leather; Methods of Sampling and Testing.
QQ-S-781—Steel, Strapping, Flat.
UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-P-271—Paper, Wrapping, Waterproofed Kraft.
CCC-C-429—Cloth, Osnaburg, Cotton.
CCC-C-467—Cloth, Jute (or Kenaf), Burlap.
PPP-B-636—Box, Fiberboard.
PPP-P-291—Paperboard, Wrapping, Cushioning.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-T-40625—Tubing, Bias Sewn (Burlap or Osnaburg) Cloth.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** The leather shall be made from cattlehides.

3.2 **Tannage.** The hide shall be tanned by a vegetable tanning process.

3.3 **Finish.** The leather shall be full grain. The flesh side shall be smooth and free from flesh and may be shaved or skived to secure greater uniformity of thickness. Splitting will be permitted to obtain extra light leather.

3.4 **Resistance to crocking (colorfastness to rubbing).** Resistance to crocking shall be not less than "good" when tested as specified in 4.3.

3.5 **Trim (form) and area.** The trim of the sides or backs shall be as specified by the definitions for "sides" or "backs" in definitions of KK-L-311. The area expressed in terms of square feet shall be legibly marked on the butt area of the flesh side of the leather.

3.6 **Color.** The color of the leather shall be as specified (see 6.2).

3.7 **Physical requirements.**

3.7.1 *Weight and thickness.* The leather shall be of the weight and thickness as specified (see 6.2) and as shown in table I, when tested as specified in 4.3.

3.7.2 *Breaking strength.* Not less than 80 percent of the specimens tested shall have minimum breaking strength values as specified in table I, when tested as specified in 4.3.

TABLE I. *Thickness and breaking strength*

Weight	Thickness (64ths of an inch)	Breaking strength, (pounds)
Extra light	8 and below 10	200
Light	10 and below 12	250
Medium	12 and below 14	300
Heavy	14 and below 16	350
Extra heavy	16 and above	400

3.7.3 *Elongation.* Not less than 80 percent of the specimens tested shall have elongation values not greater than 20 percent when tested as specified in 4.3 with the applicable load for the specified thickness indicated below:

Thickness (64ths of an inch)	Load (pounds)
8 and below 10	150
10 and below 12	200
12 and below 14	250
14 and below 16	300
16 and above	300

3.7.4 *Cracking.* The leather shall not crack when tested as specified in 4.3.

3.8 **Chemical requirements.**

3.8.1 *Class* 1. When class 1 leather is specified, the leather shall be treated with the fungicide paranitrophenol. It shall contain 0.18 to 0.70 percent fungicide, when tested as specified in 4.3.

3.8.2 The leather shall conform to the chemical requirements specified in table II, when tested as specified in 4.3.

TABLE II. *Chemical requirements*

Characteristics	Minimum	Maximum
Chloroform extract (percent)	5.0	12.0
Water-soluble material (percent)	—	25.0
Total ash (percent)	—	5.0
Degree of tannage	40.0	70.0
Acidity (pH)	3.3	4.5

3.9 **Workmanship.** The finished leather shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQLs).

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 *Examination of the unit.* The classification of defects found during the examination shall be in accordance with 4.2.1.1 and 4.2.1.2. The sample unit for these examinations shall be one side or back. The lot size shall be expressed in terms of one side or back. The inspection level and AQLs for these examinations shall be in accordance with 4.2.1.3.

4.2.1.1 *Examination for visual characteristics.*

Examine	Defect
Material	Not as specified.
Finish	Not full grain.
	Flesh side not smooth or contains areas of coarse fiber or loose flesh in excess of 4 square inches in total area.
Quality of leather	Deep cuts, open scratches, damaged grain, open grub holes, brands, hip marks, rough shoulders, or other defects which affect cutting value of more than 10 percent of the area of the side or back.
Trim	Not as specified, i.e., not side or back.
Color	Not as specified.

Note: Light, well healed scratches or grub holes, light fat wrinkles, or slight salt stains shall not be classified as defects.

4.2.1.2 *Examination for dimensional characteristics.*

Examine	Defect
Area (expressed in terms of square feet).	Not as marked on the side or back.

4.2.1.3 *Inspection level and AQL.* The inspection levels and AQLs expressed in defects per hundred units shall be as follows:

Examination	Inspection level	AQL
For defects applicable to 4.2.1.1, visual characteristics.	II	4.0
For defects applicable to 4.2.1.2, dimensional characteristics.	S-3	4.0

4.2.2 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or not as specified.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete wrapping and inadequate securing of rolls, incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
	Incorrect closure of cloth covered bundles, insecurely sewn bundles, affecting serviceability.

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Examine	Defect
Content	Not rolled grain out in set of 3 for sides and 6 for backs and the outerpiece not rolled flesh side out.
Weight	Net weight exceeds requirement.

4.3 **Tests.** The methods of testing specified in KK-L-311 wherever applicable and as listed in table III shall be followed. Sampling procedure and location from which the test unit is to be obtained shall be in accordance with KK-L-311. The specimen for testing strength requirements shall be selected from the corner of the test sample unit which is closest to the first "A" in the word "AREA". When chemical requirements are specified on a moisture-free basis, the method for moisture determination shall be method 6221. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any of the following conditions exist:

(*a*) More than 3 test failures occur for either the breaking strength or elongation requirements.

(*b*) More than 1 test failure occurs for the remaining requirements applicable to the sample unit.

(*c*) Any composite sample which fails to meet the specified requirement.

TABLE III. *Tests*

Characteristics	Requirement paragraph	Test method	Requirement applicable to:	
			Sample unit	Composite
Material identification	3.1	[1]	X	—
Tannage	3.2	[1]	X	—
Resistance to crocking	3.4	3011	X	—
Weight and thickness	3.7.1	1011[2]	X	—
Breaking strength	3.7.2	4.3.1	X	—
Elongation	3.7.3	4.3.1	X	—
Cracking	3.7.4	4.3.2	X	—
Paranitrophenol	3.8.1	4.3.3	—	X
Chloroform extract	3.8.2	6311	—	X
Water-soluble materials	3.8.2	6341[3]	—	X
Total ash	3.8.2	6421[3]	—	X
Degree of tannage	3.8.2	6641	—	X
Acidity (pH)	3.8.2	6621	—	X

[1] The supplier shall submit a certificate of compliance for these characteristics.

[2] *Note.* Method 1011 of KK-L-311 shall be used for thickness determination except that the presser foot of the spring gage shall be snapped to contact the surface of the leather.

[3] Calculated to moisture free basis.

4.3.1 *Breaking strength and elongation.* The leather shall be tested for breaking strength and elongation in accordance with methods 2011 and 2021, respectively. The elongation shall be measured simultaneously on the same specimen used for breaking strength and without stopping the travel of the jaws. The elongation shall be taken at the instant the recorder indicates the load specified in 3.7 for each respective thickness of leather. The elongation shall be recorded to the nearest 1.0 percent.

4.3.2 *Cracking.* Method 4011 shall be used except that the results shall be reported as "pass" or "fail". The diameter of the mandrel used shall be as follows:

Weight of leather	Diameter of mandrel
Extra light	5/16 inch
Light	6/16 inch
Medium	7/16 inch
Heavy	8/16 inch
Extra heavy	9/16 inch

4.3.3 *Test for the determination of paranitrophenol.* Method 6711 shall be used except that method 6321 shall not be used for the extraction procedure. If the concentration of the developed paranitrophenol color is too intense for instrument readings, the 2 ml. of aliquot shall be diluted until a reading can be taken. To compute the percent paranitrophenol use the following formula:

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$$\text{Percent paranitrophenol} = \frac{\text{Milligrams paranitrophenol in aliquot} \times \text{dilution of chloroform solution}}{10 \times (\text{weight of specimen minus moisture content in grams})}$$

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* Leather of one type, weight, class, and color only shall be tightly rolled together in multiple sets of three for sides and six for backs. When a roll of leather consists of more than one set, each additional set shall be alternated so that the heads of one set are adjacent to the butts of the next set. The leather shall be rolled grain side out except that the outer piece shall be rolled flesh side out. Each roll of leather shall be completely wrapped in 80-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. The paper shall be folded to inclose each end of the roll and shall be secured with gummed kraft paper tape. Alternatively, each roll of leather may be completely wrapped in paperboard cushioning conforming to type III, style 1 of PPP-P-291. The wrap shall be folded over to completely inclose each end of the roll. The wrap shall be secured in place by tying with jute or cotton cord or twine, applied twice around the body, approximately 12 inches from each end (when applicable) and once lengthwise, tying with nonslip knots.

5.1.2 *Level C.* Leather shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Each roll of leather, packaged as specified in 5.1, shall be completely inclosed in waterproofed kraft paper conforming to class B-2 of UU-P-271. The paper inclosing each end shall be tightly folded in place so that the roll can be inserted into cloth covering material. The waterproofed roll of leather shall be baled in bias-sewn tubing or a bag (presewn one end) made from cotton osnaburg cloth conforming to CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The bias-sewn tubing specified shall conform to MIL-T-40625. Bales shall have the open ends closed with two double-looped wire ties. The wire ties shall be not less than 6 inches long, of 0.072-inch thick galvanized soft iron or steel wire with a 1/2-inch diameter formed eye at each end. The first wire tie shall be applied as close to the bale as possible so there will be a minimum of slack in the tube, the cloth being gathered evenly to form an ear, and all edges of the cloth being 3 to 5 inches from the wire tie at the base of the ear. The second wire tie shall be applied at a distance approximately 1 inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions. Each bale shall be reinforced with two flat steel straps conforming to type I, class A, 5/8-inch wide by 0.015 inch thick of QQ-S-781. The straps shall be applied around the bale placed approximately 10 inches from each end. Sufficient tension shall be applied to the strapping, insuring tight application. Alternatively, each waterproofed roll may be packed in a snug-fitting fiberboard shipping container conforming to style FOL, V3c or V3s of PPP-P-636. Each fiberboard shipping container shall be closed, waterproofed, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Weight of the contents of each bale shall not exceed 100 pounds with a 10-pound plus tolerance permitted, and for fiberboard shipping containers the weight limitations shall be in accordance with the requirements of the container specification.

5.2.2 *Level B.* Each roll of leather, packaged as specified in 5.1, shall be baled in

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bias-sewn tubing or a bag (presewn one end) made from cotton osnaburg cloth conforming to CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The bias-sewn tubing specified shall conform to MIL-T-40625. Bales shall have the open ends closed with two double-looped wire ties. The wire ties shall be not less than 6 inches long, of 0.072-inch thick galvanized soft iron or steel wire with a 1/2-inch diameter formed eye at each end. The first wire tie shall be applied as close to the bale as possible so there will be a minimum of slack in the tube, the cloth being gathered evenly to form an ear, and all edges of the cloth being 3 to 5 inches from the wire tie at the base of the ear. The second wire tie shall be applied at a distance approximately 1 inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions. Each bale shall be reinforced with two flat steel straps conforming to type I, class A, 5/8 inch wide by 0.015 inch thick of QQ-S-781. The straps shall be applied around the bale approximately 10 inches from each end. Sufficient tension shall be applied to the strapping insuring tight application. Alternatively, each roll may be packed in a snug-fitting fiberboard shipping container conforming to style OSC (overlap not less than 3 inches), type CF, variety SW, or type SF, class domestic, of PPP-P-636. Each fiberboard shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Weight of the contents of each bale shall not exceed 100 pounds with a 10-pound plus tolerance permitted, and for fiberboard shipping containers the weight limitations shall be in accordance with the requirements of the container specification.

5.2.3 *Level C.* Leather, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The leather covered by this specification is primarily used for saddle trimming, wallets, handicraft, and orthopedic purposes.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class required (see 1.2.1).
(*c*) Color required (see 3.6).
(*d*) Weight and thickness required (see 3.7.1).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 **Civil agency procurement.** When level B packaging is specified for civil agency procurement, the requirements in 5.1.1 shall apply.

6.4 **Transportation descriptions.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Leather, cattlehide in the original tannery shapes.
Carload minimum weight: 30,000 pounds, subject to Rule 34, Uniform Freight Classification.

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Motor:

Leather, cattlehide, in the original tannery shapes.
Truckload minimum weight: 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:

Army—GL
Navy—SA
Air Force—11

Review activities:

Army—MD
Air Force—11, 67

Preparing activity:

Army—GL

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